

Arrow Lake operator stations: Mesa fix

TechOps support runbook · TEC-880 · tool: `techops-tools/operator_station/arrow-lake-mesa/arrow-lake-mesa.sh` · October 2026

In one paragraph. Operator stations with an Intel **Arrow Lake** GPU (PCI id `8086:7d67`) run our Ubuntu 22.04 image with Mesa 23.2.1, which does not support that GPU. Chrome and the desktop then draw everything on the CPU. The station runs close to its limit even when idle, and under load the screens freeze or lag and video stalls. **Rebooting does not help.** Installing a newer Mesa (25.x) moves the drawing back to the GPU. It needs no hardware change and no Ubuntu upgrade, and can be rolled back.

1. Is this station affected?

Signs on a station that needs the fix:

- Frozen or slow screens, stuck video, or the kiosk relauching, often on **busy, multi-screen** stations.
- `lspci` shows `8086:7d67` , and Mesa is `23.2.1` .
- The desktop compositor (`gnome-shell`) uses ~70–80% of a CPU core, and CPU pressure is 4–18% at idle.

Fleet scan, 4–5 Oct 2026 (stations reachable over SSH):

Status	Stations
Affected, not fixed (13)	kela-fob-04 (17.6% pressure, heaviest), fob-05, fob-08, fob-10, fob-11, fob-20, ogmar80 (both stations), palmam-bolem-103, palmam-bolem-104, palmam-einav, palmam-golan, palmam-negev
Fixed with Mesa 25 (3)	kela-tlv-dev-03 (pilot), kela-fob-06, kela-fob-21
Known Arrow Lake from tickets	kela-kanaf / kanaf-operator-3, afbp-ovda
Not affected: older GPU <code>8086:a780</code> , driver OK	kela-fob-12, 14, 15, 16, 17, 18, 19, 27
Not checked yet	kela-fob-01, 03, 07, 09, fob-96-hohit-hub, bolem-103 rugged station

Priority, by complaint count since July: **fob-08** (14 tickets), **fob-10** (11), **fob-04** (8), then the rest. fob-21 (9) is done. Arrow Lake sites file ~3x more screen and video tickets per site than sites with a working driver.

2. Before you start

- **Coordinate with the site.** The final step reboots the station: **all its screens go dark for ~2–3 minutes**. Pick a quiet moment and tell the position.
- You need SSH to `KelaAdmin@<site>-operator` (over Tailscale) and the KelaAdmin station password. The tool never stores or passes it; ssh and sudo ask for it, 2–3 times per run.
 - **Paste the password, don't type it**, and make sure the keyboard is on **English**. A Hebrew layout makes every character wrong. Typing `y` on Hebrew gives `ו` .
 - If the 1Password value is rejected, use the station password Tobee uses (known to work fleet-wide):
`ssh -t -i ~/.ssh/tobee.pem ubuntu@tobee "sudo grep '^TOBEE_OPERATOR_PASSWORD=' /opt/tobee/.env | cut -d= -f2-"` . After several wrong tries the station may refuse logins for a minute or so.

- Get the tool: `git clone / git pull` the `techops-tools` repo, then `cd operator_station/arrow-lake-mesa` and **run every command from that folder**. That's where it keeps the before/after snapshots that `verify` compares. It runs on macOS or Linux (bash, ssh).
- If ssh warns **REMOTE HOST IDENTIFICATION HAS CHANGED**, the station was reinstalled. Don't bypass it on your own; ask TechOps to confirm the new fingerprint first. Then `ssh-keygen -R <station>` and reconnect.
- Slow sites download ~27 MB: fob-21 took ~9 minutes at 52 kB/s. That's normal, not a hang.

3. The fix: three commands

Step 1: snapshot before (read-only, optional but recommended)

```
./arrow-lake-mesa.sh check kela-fob-08-operator
```

Prints the GPU, Mesa version, CPU pressure, compositor/Xorg/Chrome CPU and Chrome's own GPU report, and saves it under `./arrow-lake-reports/`. The last line tells you whether the station is a candidate.

Step 2: apply (installs Mesa, then asks before rebooting)

```
./arrow-lake-mesa.sh apply kela-fob-08-operator
```

1. Saves a **before** snapshot automatically.
2. Refuses unless the GPU is `8086:7d67`, the OS is 22.04 and the kernel driver is `i915`. It does nothing if the station is already fixed.
3. Asks for the KelaAdmin password (sudo), adds `ppa:kisak/turtle` and upgrades the Mesa packages. This runs non-interactively, so the "Daemons using outdated libraries" dialog does not appear.
4. Asks *"Reboot now? [y/N]"*. Answer **y** when the site is ready.

Step 3: verify (~5 minutes after the reboot)

```
./arrow-lake-mesa.sh verify kela-fob-08-operator
```

Takes an **after** snapshot, prints it next to the **before**, and checks:

Check	Pass when
Mesa	version contains <code>kisak</code> (e.g. <code>25.0.7~kisak3~j</code>)
Chrome renderer	contains <code>Mesa Intel(R) Graphics (ARL)</code> (not <code>llvmpipe / SwiftShader</code>)
GPU compositing	<code>enabled</code>
Desktop compositor (gnome-shell)	under 20% CPU (was ~70–80%)
Kiosk	at least one C2 tab is open

It ends with **RESULT: PASS** or **RESULT: FAIL**. If the station has been up under 5 minutes it says so; run `verify` again a few minutes later.

4. What good looks like

kela-tlv-dev-03 (1 screen)			kela-fob-06 (3 screens, 7 streams)		
	Before	After		Before	After
gnome-shell	81%	3%	gnome-shell	76%	4%
Xorg	16%	3–4%	Xorg	20%	<5%
Chrome GPU process	58%	30%	CPU pressure	13–14%	~5.5%
Load	3.9	~1.7–2	Chrome (all)	~163%	~172%
Renderer	software	Mesa Intel (ARL)	Load	5.5	5.2

kela-fob-21 (2 screens), 40 min after the fix: gnome-shell **3.4%**, Xorg **2.5%**, CPU pressure **3.5%**, both screens on C2, no kiosk relaunches.

Expected limit: the drawing load disappears, but **video decoding stays on the CPU**, because there is no Arrow Lake video driver for 22.04. That is why Chrome's own CPU barely changes on busy multi-screen stations. Hardware video decode comes with the 24.04 image (TEC-910). On the heaviest stations, also reduce load where the site agrees: force 1080p (`KIOSK_PREFERRED_RESOLUTION`), use the 2D map, keep one screen on camera, and disconnect unused screens.

5. If something goes wrong: rollback

```
./arrow-lake-mesa.sh rollback kela-fob-08-operator
```

Removes the PPA, returns to Ubuntu's own Mesa 23.2.1 with `ppa-purge` , and asks before rebooting. Use it if the screens come back black, the kiosk does not start, or `verify` fails twice.

6. Doing it by hand (if the tool can't be used)

The same steps, run from your laptop. Each asks for the KelaAdmin password.

Before: read-only metrics

```
ssh KelaAdmin@<station> 'lspci -nn | grep -i vga; lspci -k -s 00:02.0 | grep "driver in use"; dpkg-query -W libgl1-mesa-dri; head -1 /proc/pressure/cpu; uptime; top -b -n 2 -d 5 -w 200 | awk "/^top -/{n++} n==2" | grep -E "gnome-shell|Xorg|chrome" | head'
```

Install

```
ssh -t KelaAdmin@<station> 'sudo add-apt-repository -y ppa:kisak/turtle && sudo apt-get update -q && sudo env DEBIAN_FRONTEND=noninteractive NEEDRESTART_MODE=l apt-get install -y -q --only-upgrade libgl1-mesa-dri libglx-mesa0 libegl-mesa0 libgbm1 mesa-vulkan-drivers'
```

One command on one line. Copy it whole.
If a "Daemons using outdated libraries" dialog still appears: leave only `fwupd` ticked and press Enter on **<Ok>**. **Do not** tick `gdm3` (it would log the operator out mid-install). A line saying `packagekit.service` is masked is harmless.

Reboot (coordinate first)

```
ssh -t KelaAdmin@<station> 'sudo reboot'
```

After (~5 min): same read-only metrics as "Before". Success means the new Mesa version, gnome-shell well under 20%, and Xorg a few percent.

Chrome's own view: on the station screen (AnyDesk) open `chrome://gpu` . *Compositing* and *Rasterization* should say **Hardware accelerated**, and the GL renderer should be "Mesa Intel(R) Graphics (ARL)".

Rollback

```
ssh -t KelaAdmin@<station> 'sudo env DEBIAN_FRONTEND=noninteractive NEEDRESTART_MODE=l apt-get install -y -q ppa-purge && sudo env DEBIAN_FRONTEND=noninteractive NEEDRESTART_MODE=l ppa-purge -y ppa:kisak/turtle'
```

Then reboot, after telling the site, with the same command as above: `ssh -t KelaAdmin@<station> 'sudo reboot'` .

7. Troubleshooting

Symptom	What to do
<code>apply</code> refuses: GPU not 7d67	Not an Arrow Lake station. This fix does not apply; follow the operator-station runbook.
<code>apply</code> refuses: driver not i915	The kernel isn't driving the GPU. Don't apply; escalate to TechOps (it may need <code>i915.force_probe=7d67</code>).
Screens black after the reboot	Wait 5 minutes. If still black: <code>rollback</code> , then escalate.
<code>verify</code> can't read Chrome's GPU report	The kiosk is not up yet, or Chrome's DevTools port is off. Wait and re-run, or check <code>chrome://gpu</code> on screen.
Video tiles drop all at once and C2 shows its connection error, then recovers	A different problem: the station loses its connection to the server (network/cable/switch). Mesa does not fix it. Report it separately with the time it happened.
Still slow on a 3–4 screen station after PASS	Expected video-decode load (see §4). Apply the load reductions and report the station.

8. After the fix

- Add a note to the ticket: station, time, Mesa version, and the `verify` result (paste the output or attach the files from `arrow-lake-reports/`).
- The incident-agent's station check still flags "Arrow Lake, no GPU driver" by GPU model alone, until it learns to read the Mesa version. Ignore that flag on fixed stations.
- Fleet tracking: TEC-880 (this fix), TEC-910 (24.04 image, where video decode also gets fixed).